

Sports Platform - Full End-to-End Mermaid Architecture

1. User Onboarding

```
sequenceDiagram
    participant User
    participant MobileApp
    participant APIGateway
    participant AuthService
    participant ProfileService
    participant UserDB
    User->>MobileApp: Sign Up
    MobileApp->>APIGateway: Register
    APIGateway->>AuthService: Create Account
    AuthService->>UserDB: Save User
    AuthService-->>APIGateway: Success
    APIGateway->>ProfileService: Create Profile
    ProfileService-->>APIGateway: Profile Created
    APIGateway-->>MobileApp: Success
```

2. Ground Booking

sequenceDiagram

participant User

participant APIGateway

participant VenueService

participant BookingService

participant PaymentService

participant BookingDB

participant RabbitMQ

User->>APIGateway: Book Ground

APIGateway->>VenueService: Check Availability

VenueService-->>APIGateway: Available

APIGateway->>BookingService: Create Booking

BookingService->>PaymentService: Pay

PaymentService-->>BookingService: Success

BookingService->>BookingDB: Store Booking

BookingService->>RabbitMQ: BookingCreated

3. Booking Cancellation

```
sequenceDiagram
    participant User
    participant BookingService
    participant PaymentService
    participant VenueService
    participant NotificationService
    User->>BookingService: Cancel Booking
    BookingService->>PaymentService: Refund
    PaymentService-->>BookingService: Success
    BookingService->>VenueService: Release Slot
    BookingService->>NotificationService: Notify User
```

4. Tournament Registration

sequenceDiagram

participant User

participant TournamentService

participant PaymentService

participant AnalyticsService

participant NotificationService

User->>TournamentService: Register

TournamentService->>PaymentService: Fee Payment

PaymentService-->>TournamentService: Success

TournamentService->>AnalyticsService: Update Stats

TournamentService->>NotificationService: Send Ticket

5. Match Creation

```
sequenceDiagram
    participant User
    participant MatchService
    participant PlayerService
    participant NotificationService
    User->>MatchService: Create Match
    MatchService->>PlayerService: Find Players
    PlayerService-->>MatchService: Players
    MatchService->>NotificationService: Send Invites
```

6. Player Discovery

sequenceDiagram

participant User

participant SearchService

participant PlayerService

participant RecommendationService

User->>SearchService: Find Players

SearchService->>PlayerService: Nearby Players

SearchService->>RecommendationService: Rank Players

RecommendationService-->>SearchService: Results

7. Smartwatch Sync

sequenceDiagram
participant SamsungHealth
participant WearableService
participant FitnessDB
participant AnalyticsService
SamsungHealth->>WearableService: Sync Metrics
WearableService->>FitnessDB: Save Data
WearableService->>AnalyticsService: Analyze Fitness

8. Recommendation Engine

```
sequenceDiagram
    participant User
    participant RecommendationService
    participant AnalyticsService
    participant VenueService
    participant TournamentService
    RecommendationService->>AnalyticsService: User History
    RecommendationService->>VenueService: Nearby Venues
    RecommendationService->>TournamentService: Events
    RecommendationService-->>User: Recommendations
```


9. Ground Owner Flow

```
sequenceDiagram
    participant Owner
    participant VenueService
    participant VerificationService
    participant AnalyticsService
    Owner->>VenueService: Register Ground
    VenueService->>VerificationService: Verify Docs
    VerificationService-->>VenueService: Approved
    VenueService->>AnalyticsService: Track Revenue
```

10. Organizer Flow

sequenceDiagram

participant Organizer

participant TournamentService

participant PaymentService

participant NotificationService

Organizer->>TournamentService: Create Event

TournamentService->>PaymentService: Fee

TournamentService->>NotificationService: Publish Event

11. Analytics Pipeline

```
sequenceDiagram
    participant Services
    participant RabbitMQ
    participant AnalyticsService
    participant AnalyticsDB
    Services->>RabbitMQ: Events
    RabbitMQ->>AnalyticsService: Consume
    AnalyticsService->>AnalyticsDB: Store Metrics
```

12. End to End Journey

sequenceDiagram

participant User

participant MobileApp

participant APIGateway

participant Services

User->>MobileApp: Signup

User->>MobileApp: Connect Watch

User->>MobileApp: Book Ground

User->>MobileApp: Create Match

User->>MobileApp: Join Tournament

Services-->>User: Analytics & Insights